
Bottleneck-Typed Retrieval for Legal and Multi-Hop Question Answering

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Abstract

RAG should be selected as a bottleneck-targeted intervention: top- k sensitivity diagnoses whether the active failure is retrieval depth, query formulation, evidence use, or answer-option anchoring. We began with a full LangGraph legal RAG agent, but exploratory results showed that extra planning and retrieval could distract models with strong legal priors. We therefore reframed the project around a bottleneck-typed evaluation across BarExam, MuSiQue, CaseHOLD, LegalBench-SCALR, and HousingQA. The same retrieval-depth policy stress test separates regimes. On Llama 70B MuSiQue, restricting rag_simple from top-5 to top-1 passages drops EM from 27.5% to 13.0% (-14.5pp, McNemar $p = 4.18 \times 10^{-7}$). On an N=200 Gemma 4 26B BarExam diagnostic, the analogous top-5/top-1 comparison is flat (82.5% to 83.0%, $p = 1.00$). LegalBench-SCALR is candidate-depth sensitive (77.0% to 59.5% at top-1), while HousingQA is a promising but directional statutory replicate (50.5% top-1 to 58.0% top-10, $p = 0.0722$). Snap-HyDE is used as a probe, not the novelty claim: on MuSiQue it improves Llama 70B from 27.5% to 37.0% (+9.5pp, $p = 0.0079$), while BarExam’s full-corpus Gemma 4 26B rag_snap_hyde result improves from 78.08% to 81.17% ($N = 1195$). The central lesson is not that one RAG method wins, but that retrieval methods should be chosen according to the benchmark-model bottleneck.

1 Introduction

Legal question answering is a difficult testbed for RAG. The model must apply rules, exceptions, and fact patterns precisely, and the retrieved text may be only partially aligned with the issue in the question. Our original project therefore built an agentic legal RAG system: a classifier/planner decomposed a question, an executor retrieved and synthesized evidence for each step, an evaluator judged whether evidence was sufficient, and a replanner added or revised steps when the evidence was incomplete.

The first lesson was negative but useful. In exploratory BarExam checks, the full agentic loop was expensive and did not reliably outperform direct model answering. We treat those early rows as motivation rather than signed-off evidence. They exposed the broader problem that more retrieval and more agent structure can distract a model that already has strong priors, while also fragmenting reasoning across multiple calls.

We pivoted from “build the largest agent” to “measure which bottleneck is active.” The result is a bottleneck-typed account of RAG. Some benchmark-model pairs are *retrieval-depth limited*: answer

accuracy is sensitive to whether enough useful passages are retrieved. Others are *depth-flat*: the useful intervention is answer anchoring or option calibration, not more documents. Holding-selection tasks split further: SCALR is candidate-depth limited until top-5, while CaseHOLD is answer-flat under the current harness and still needs repaired retrieval-recall instrumentation.

This framing also explains why a seemingly weak control, `golden_passage`, can underperform `llm_only` and `rag_snap_hyde` on BarExam. Providing a single labeled passage is not an oracle ceiling. In our full Gemma 4 26B BarExam audit, `golden_passage` flips 96 previously correct `llm_only` answers to wrong and only 83 wrong answers to correct. The gold passage often anchors the model on a related but incomplete rule. In contrast, the pattern is consistent with snap-first methods preserving useful legal priors before evidence reconciliation, which is why `snap_only_in_final` and `rag_snap_hyde` both beat the forced gold-passage control.

2 Related Work

Our method is related to generated-query and draft-conditioned retrieval. HyDE generates a hypothetical document and embeds it for zero-shot dense retrieval [1]. Query2Doc similarly uses LLM-generated pseudo documents for query expansion [2]. FLARE retrieves using tentative generated text during long-form generation [3], while Iter-RetGen and CoRAG iterate between generation and retrieval [4, 5]. Speculative RAG uses multiple retrieved document subsets to draft answers and a verifier to select among them [6]. Our snap-HyDE variant is simpler and fixed-cost: one call produces a tentative answer and a retrieval passage, and a second call synthesizes the final response.

The more important distinction is the bottleneck framing. Adaptive-RAG routes questions by complexity [7]; Know Your RAG and GRADE classify question difficulty and multi-hop structure [8, 9]. Legal RAG benchmarks have already shown that retrieval helps some legal tasks and hurts others. Zheng et al. introduce BarExam QA and HousingQA and report that retrieval helps Housing much more than BarExam [10]. Vaddi reports task-dependent legal RAG behavior, including gains on ContractNLI and degradation on CaseHOLD [11]. We build on these observations by treating retrieval benefit as a measurable bottleneck signature rather than a universal property of the retriever.

This boundary matters. We do not claim that snap-HyDE is fundamentally novel relative to HyDE, generated-query retrieval, or active retrieval. Instead, the contribution is a diagnostic matrix: no-context, top-1/top-5 retrieval, single gold-passage controls, generated-query methods, and snap-anchored variants together identify whether a benchmark is retrieval-depth limited, query-formulation limited, evidence-utilization limited, or answer-option disambiguation limited.

3 System and Method

3.1 Initial agentic legal RAG system

The initial system was a LangGraph plan-execute-evaluate loop. A planner constructed a legal research plan, an executor ran multi-query retrieval and source-grounded synthesis for each step, a judge checked whether the retrieved evidence was sufficient, and a replanner added follow-up research steps when needed. This design was useful for exploring legal-agent architecture, but the evaluation showed that its many moving pieces made it hard to identify where accuracy gains or losses came from.

3.2 Snap-conditioned retrieval

The core method family is snap-conditioned retrieval. In the legal multiple choice setting, the model first produces a tentative answer or legal analysis. That snap is then used to guide a HyDE-style passage or evidence query. The retriever searches with this answer-shaped query, and a final synthesis call combines the original question, snap, and retrieved passages.

The two-call version, `snap_hyde_2call`, makes this fixed-cost:

1. Call 1: produce a tentative answer plus a reference passage stating the controlling rule, doctrine, fact, or principle.
2. Retrieval: embed the generated passage and retrieve the top- k passages from the corpus.

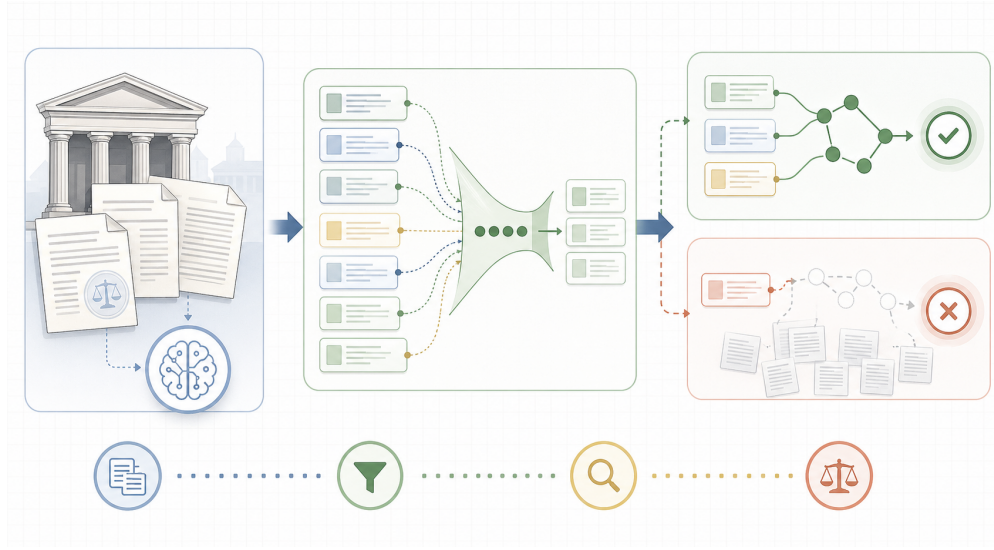


Figure 1: Project pivot. The initial architecture emphasized a full legal research agent. The final experiments sort questions through controlled retrieval interventions and ask which bottleneck produces success or failure.

3. Call 2: synthesize the final answer using the question, snap, and retrieved evidence.

This design is not claimed as a universal invention independent of prior work. Its value in this project is as a compact probe: when it helps, we can ask whether the gain came from better retrieval recall, stronger answer anchoring, or both.

3.3 Bottleneck diagnostic

We use retrieval-depth sensitivity as a simple diagnostic. If reducing `rag_simple` from top-5 to top-1 retrieved passages sharply reduces accuracy, the benchmark-model pair is retrieval-bottlenecked. If the result is flat, additional retrieval depth is not the active bottleneck. This diagnostic is method-agnostic as a first pass because it can be applied before testing more elaborate retrieval shaping. It is still a retrieval-policy stress test, not pure final-context truncation: in our harness, the cross-encoder reranker selects the final top- k passages from a first-stage pool of roughly $3k$ candidates.

4 Experimental Setup

Datasets. For answer-quality experiments, we evaluate on BarExam QA, MuSiQue, CaseHOLD, LegalBench-SCALR, and HousingQA. BarExam is a legal multiple-choice benchmark over bar-exam fact patterns. MuSiQue is open-ended multi-hop QA. CaseHOLD and SCALR test holding selection among candidate legal holdings. HousingQA tests yes/no statutory reasoning over state housing law. We also use MLEB-SCALR as a retrieval-only calibration benchmark with qrels, not as a headline answer-accuracy result.

Models. The main legal full-corpus results use Gemma 4 26B-A4B and Gemma 4 E4B served through the cluster evaluation path. The main multi-hop results use Llama 3.3 70B. We also use Gemma 3 27B as an important negative cross-family check on MuSiQue.

Evaluation. For answer accuracy, we report exact match or multiple-choice accuracy depending on the dataset. For paired comparisons, we use McNemar tests on the same question slice. Retrieval-only MLEB-SCALR metrics were computed as side calibration, not as answer-quality evidence. We treat small samples as directional and emphasize full-corpus or paired $N = 200$ results when drawing conclusions.

5 Results

5.1 BarExam: snap-first reasoning beats raw retrieval and gold context

Table 1: BarExam Tier 3 method matrix: Gemma 4 26B-A4B, $N = 1195$.

Method	Accuracy	Δ vs rag_simple
rag_simple	78.08%	–
golden_passage	78.66%	+0.58pp
rag_hyde	78.91%	+0.83pp
llm_only	79.75%	+1.67pp
snap_only_in_final	80.59%	+2.51pp
rag_snap_hyde	81.17%	+3.09pp

The BarExam result is the cleanest full-corpus legal finding. On Gemma 4 26B-A4B, rag_snap_hyde reaches 81.17%, beating rag_simple by +3.09pp (paired b/c=124/87, $p = 0.0130$). The same winner appears on Gemma 4 E4B, improving from 58.49% to 62.18% (+3.68pp, b/c=172/128, $p = 0.0129$). Figure 2 shows the cross-size pattern. The row is approved with a minor caveat: the audited rag_simple baseline included 2/15 sampled startup null predictions with empty retrieval, while the headline re-scored totals still hold.

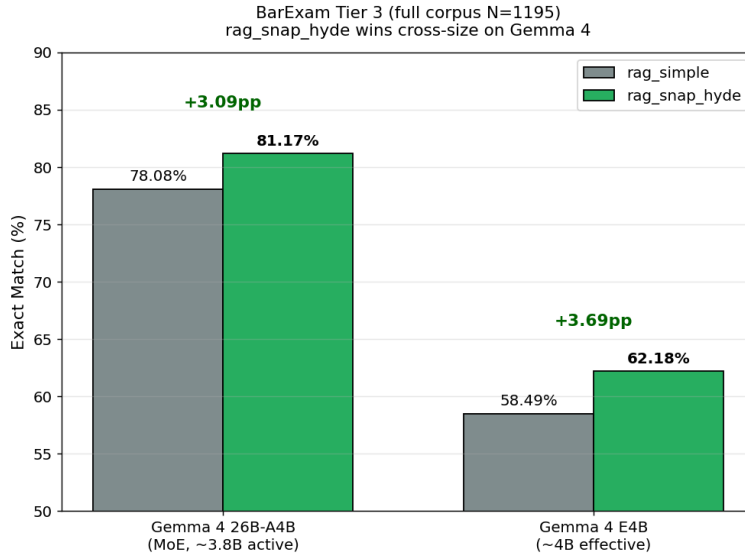


Figure 2: BarExam full-corpus cross-size result. rag_snap_hyde improves over rag_simple on both Gemma 4 model sizes.

The important mechanism is that BarExam is not simply “retrieval helps.” In fact, llm_only beats golden_passage, and snap_only_in_final is close to the best method. Our audit showed that gold-passages injection is a noisy single-snippet control: it helps some cases but also anchors the model to partial or misleading doctrine. The snap-first methods work because they preserve the model’s strong legal prior before asking it to reconcile with retrieved text.

5.2 MuSiQue: retrieval depth is first-order

MuSiQue behaves differently. The Llama 70B baseline rag_simple achieves 27.5% EM. snap_hyde_2call reaches 37.0%, a +9.5pp improvement with paired McNemar $p = 0.0079$. multi_hyde_diverse and iterative_planning_table also improve substantially, while the current subagent_rag implementation is significantly worse because its gap-routing prompts over-abstain.

Table 2: Llama 70B MuSiQue paired results, $N = 200$.

Method	EM	Δ vs rag_simple	McNemar p
rag_simple	27.5%	–	–
rag_multi_query	29.0%	+1.5pp	0.728
multi_hyde_diverse	35.5%	+8.0pp	0.0195
iterative_planning_table	36.0%	+8.5pp	0.0533
snap_hyde_2call	37.0%	+9.5pp	0.0079
subagent_rag	15.5%	-12.0pp	0.0007

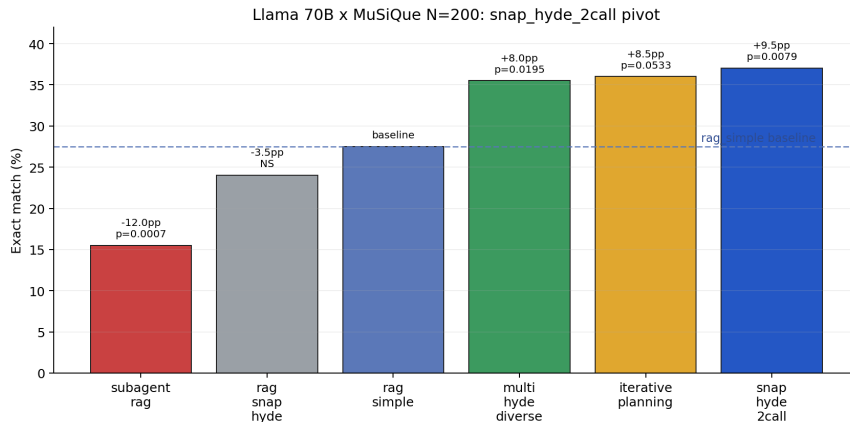


Figure 3: Current MuSiQue pivot for Llama 70B, $N = 200$. The two-call snap-HyDE variant has the best point estimate and is significant versus rag_simple; it is not statistically separated from multi_hyde_diverse or iterative_planning_table.

The retrieval-depth ablation shows why this is a retrieval-bottlenecked regime. Restricting rag_simple from top-5 to top-1 passages drops EM from 27.5% to 13.0% (-14.5pp, $p = 4.18 \times 10^{-7}$). Across MuSiQue methods, top-1 causes large drops of roughly 10–17pp. Passage count is therefore a first-order variable; method choice matters, but only after the system retrieves enough evidence.

5.3 Cross-dataset bottleneck taxonomy

Table 3: Retrieval-depth and method behavior across datasets.

Dataset / model	Retrieval-depth diagnostic	Bottleneck read / caveat
MuSiQue / Llama 70B (N=200)	top-5 27.5% to top-1 13.0% (-14.5pp, $p = 4.18 \times 10^{-7}$)	Retrieval-depth limited; snap_hyde_2call adds a query-formulation lift to 37.0%.
BarExam / Gemma 4 26B (N=200 diagnostic)	top-5 82.5% to top-1 83.0% (+0.5pp, $p = 1.00$)	Depth-flat; the full-corpus snap lift is modest and anchoring-driven, not more-documents-driven.
CaseHOLD / Llama 70B (N=200)	top-5 72.0% to top-1 70.5% (-1.5pp, $p = 0.678$); two-call 69.5%	Answer-flat under current harness; do not cite gold-hit until repaired CaseHOLD Chroma is rebuilt.
LegalBench-SCALR / Llama 70B (N=200)	top-5 77.0% to top-1 59.5% (-17.5pp, $p = 1.05 \times 10^{-8}$); top-10 = 77.0%	Candidate-depth limited, then saturated by top-5 despite higher top-10 gold-hit.
HousingQA / Gemma 4 26B (N=200)	top-1 50.5% to top-10 58.0% (+7.5pp, $p = 0.0722$); two-call 57.0%	Directional statutory retrieval-depth signal; state metadata and gold mapping remain caveats.

This table is the main conceptual result. Legal QA is not one regime. BarExam and CaseHOLD are answer-flat under current depth and two-call probes, while SCALR is retrieval-depth sensitive and HousingQA is a directional statutory analogue. The CaseHOLD row is a negative control, not retrieval-recall evidence: old logs report 0/200 gold-hit because the gold-option mapping was not addressable until the repair. HousingQA is now a landed directional result rather than merely a queued statutory test.

6 Mechanism Analysis

6.1 Why the gold passage can be worse than snap-HyDE

The BarExam golden-passage result initially looks paradoxical: if the system is given the gold passage, why does it underperform `llm_only` and `rag_snap_hyde`? The answer is that the gold passage is a control, not an oracle. A single passage can be topically related without containing the decisive rule, or it can emphasize a distractor doctrine that is plausible but wrong for the fact pattern.

In the paired Gemma 4 26B BarExam audit, `llm_only` gets 79.75% while `golden_passage` gets 78.66%. The transition table shows 96 `llm_only`-correct answers flipped wrong by the gold passage, compared with 83 `llm_only`-wrong answers fixed by the passage. Meanwhile, `rag_snap_hyde` reaches 81.17%. A 30-case manual audit split the paradox cases into topically irrelevant gold passages (30%), relevant-but-misapplied passages (50%), and distractor or partial-rule passages (20%). Snap anchoring is therefore helpful but not magic: `rag_snap_hyde` still follows the wrong gold-induced answer in 28 of the 96 paradox cases. The mechanism is that snap-first reasoning gives the model an initial legal answer to reconcile against evidence, rather than letting one provided snippet dominate the first read.

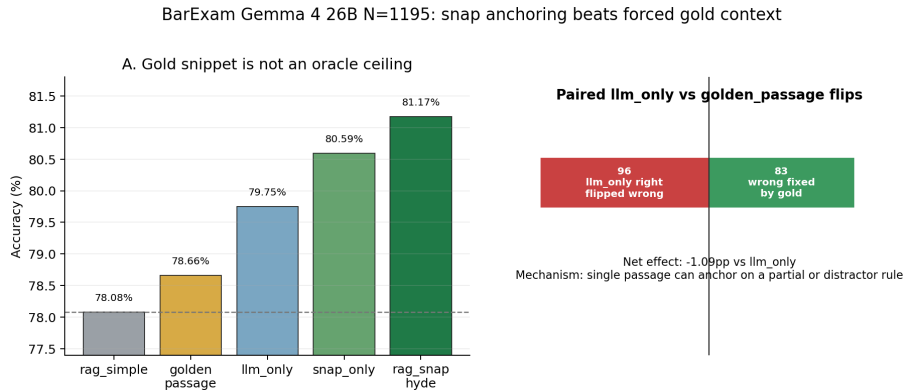


Figure 4: BarExam mechanism audit. The single gold-passage control is not an oracle ceiling: it flips more `llm_only`-correct answers wrong than wrong answers right, while snap-first methods recover some of that loss.

6.2 Gold context is task-dependent

The same control reverses on MuSiQue. In a new Llama 70B N=200 paired `golden_passage` run, providing the labeled gold passage reaches 56.5% EM, compared with 27.5% for `rag_simple` (+29.0pp, $p = 2.44 \times 10^{-13}$) and 37.0% for `snap_hyde_2call` (+19.5pp for gold over two-call, $p = 8.07 \times 10^{-8}$). This does not make `golden_passage` a deployable method; it is a control with access to the answer-bearing context. But it shows why the BarExam result is not a generic indictment of context injection. Gold context is useful when the model lacks facts and harmful or noisy when a legal MC prior is already strong and the provided snippet can anchor the wrong doctrine.

6.3 Why snap-HyDE is model-conditional on MuSiQue

The MuSiQue story is more retrieval-dominant. Llama 70B’s snap-HyDE passages improve gold-passage retrieval, and EM rises. Gemma 27B’s snap-HyDE passages move retrieval away from gold passages, and EM falls. This explains why `snap_hyde_2call` is not a model-agnostic multi-hop cure: the method helps only when the model can write a generated passage that matches the corpus style and points retrieval toward the right evidence.

Table 4: MuSiQue snap-HyDE mechanism by model, $N = 200$.

Model	RAG EM	2-call EM	RAG gold-hit	2-call gold-hit
Llama 70B	27.5%	37.0%	84.0%	86.5%
Gemma 27B	28.5%	23.0%	84.0%	76.5%

6.4 Cost and practicality

The final methods are intentionally much cheaper than the first agentic loop. The full LangGraph agent used many calls per question and made debugging difficult. `snap_hyde_2call` fixes the budget at two LLM calls per question, with about 1,384 total tokens per question. It slightly exceeds `multi_hyde_diverse` in EM at a similar two-call budget and far lower latency, while `iterative_planning_table` spends 6.8 calls and roughly 2,929 tokens per question for a similar point estimate. Figure 5 shows that the useful methods are not just the highest-accuracy methods; cost per question matters.

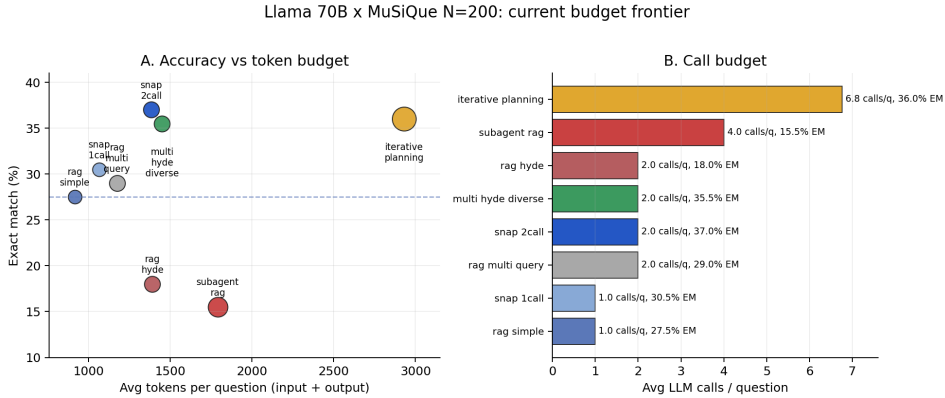


Figure 5: Current cost vs. accuracy frontier on Llama 70B MuSiQue, $N = 200$. Marker size tracks LLM calls. `snap_hyde_2call` is the best current point estimate at fixed two-call cost, while `iterative planning` spends far more calls for comparable EM.

7 Limitations

The strongest MuSiQue result is paired $N = 200$, not a full-corpus run. It is statistically significant, but a larger replicate would make the claim more robust. The BarExam result is full-corpus and cross-size, but its effect size is modest; it should be presented as a mechanism result rather than a dramatic accuracy breakthrough. CaseHOLD previously lacked meaningful gold-option retrieval instrumentation; we repaired the mapping, but new retrieval claims require a rebuilt Chroma collection and fresh logs. HousingQA has landed only as a directional $N=200$ diagnostic; the top-10 lift is suggestive, but state metadata and weak gold-hit mapping still need a cleaner causal audit.

The method is also provider- and model-dependent. OpenRouter-served Gemma runs showed runaway-loop behavior on iterative prompts, so Gemma results should prefer cluster-served vLLM logs. More broadly, snap-HyDE depends on the model’s ability to generate a useful intermediate passage. If the generated passage is off-style or wrong, retrieval can get worse.

8 Conclusion

The final project result is a more nuanced account of agentic legal RAG than the original proposal. Heavy planning agents are not automatically better than simple baselines. Retrieval is not automatically helpful, and gold context is not an oracle. The useful direction is bottleneck-aware method selection: multi-hop QA needs enough retrieved evidence and benefits from answer-shaped retrieval; BarExam-style legal MC benefits more from snap-first reasoning and careful reconciliation; holding-selection tasks need candidate-set diagnostics; statutory QA likely needs metadata-aware retrieval. This is the strongest story for a class report and the best foundation for future work on adaptive legal RAG systems.

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